

CampusWater AI

AI-Powered Water Usage Monitoring & Conservation Assistant

1M1B AI for Sustainability Virtual Internship

In Collaboration with IBM SkillsBuild & AICTE

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1. Project Description

Title

CampusWater AI - AI-Powered Water Usage Monitoring & Conservation Assistant

SDG Alignment

Primary: SDG 6 - Clean Water and Sanitation (Target 6.4: Water-use efficiency, Target 6.B: Community participation)

Secondary: SDG 11 - Sustainable Cities and Communities (Target 11.6: Environmental impact of cities)

Secondary: SDG 13 - Climate Action (Target 13.3: Climate awareness and capacity)

Problem Statement

How might we use AI to monitor, predict, and optimize water consumption patterns in college campuses so that water resources can be managed more sustainably and wastage can be reduced by at least 20%?

AI Solution Overview

CampusWater AI is a conversational AI assistant that integrates with campus water metering infrastructure to provide real-time monitoring, usage forecasting, leak detection alerts, and personalized conservation recommendations. It uses a Retrieval-Augmented Generation (RAG) pipeline with IBM Granite models to answer water-related queries and generate actionable insights for campus administrators and students alike.

Target Users

- Campus administration / facility management teams
- Hostel wardens and maintenance staff
- Students and faculty (for awareness and behavioral change)
- Local municipal water authorities (for reporting)

2. Problem Context & Real-World Relevance

Water scarcity is one of the most pressing challenges of our time. According to UN Water, 2.3 billion people live in water-stressed countries, and demand is expected to exceed supply by 40% by 2030.

College campuses are microcosms of cities - they consume significant water for hostels, cafeterias, laboratories, landscaping, and sanitation. Yet most campuses lack real-time visibility into their water usage patterns. Leaks go undetected for days, consumption spikes go unexplained, and there is no data-driven feedback loop to encourage conservation.

CampusWater AI addresses this gap by bringing AI-powered analytics to campus water management - making it accessible, affordable, and actionable.

Who is affected?

- Students in hostels face water shortages during peak hours due to uneven distribution
- Campus administration pays high utility bills without insight into wastage patterns
- Local communities downstream bear the environmental cost of excessive extraction
- Future generations inherit depleted water tables and stressed ecosystems

Why is AI needed?

- Prediction: AI can forecast water demand patterns based on historical data, weather, and academic calendar
- Anomaly Detection: Sudden spikes in usage indicate leaks - AI can flag them in real-time
- Personalization: AI tailors conservation tips to individual user behavior patterns
- Scale: AI can process data from hundreds of metering points across campus simultaneously
- Natural Language Interface: A chatbot makes data accessible to non-technical staff

3. Design Thinking Process

Stage 1: Empathize

- Interviewed hostel students: 70% reported experiencing water shortages at peak hours
- Spoke to maintenance staff: leak detection is manual and takes 2-3 days on average
- Reviewed campus water bills: 15-20% monthly variation suggests unidentified wastage
- Observation: No feedback mechanism exists to inform users about their consumption

Stage 2: Define

Problem Statement: Campus water management is reactive, not proactive. There is no real-time data, no predictive capability, and no personalized user engagement - leading to significant undetected wastage and inequitable distribution.

Stage 3: Ideate

- Real-time dashboard with AI-powered consumption forecasts
- WhatsApp/Telegram chatbot for instant water usage queries
- Leak detection alerts via automated anomaly detection
- Gamified conservation challenges with leaderboards for hostels
- Integration with IoT smart meters for automated data ingestion

Stage 4: Prototype

The prototype consists of three components:

- RAG Pipeline: IBM Granite model retrieves campus water data and generates natural-language answers
- Conversational Interface: Chatbot that accepts queries like 'What was our water usage yesterday?' or 'Are there any anomalies in Block A?'
- Alert System: Automated notifications when consumption exceeds predicted baseline by 2 standard deviations

Stage 5: Test & Refine

- Tested with 5 sample users (2 admin staff, 3 students)
- Feedback: Users wanted simpler language and more visual representation
- Refinement: Added summary cards and comparison with previous week
- Ethical check: No personal data stored - only building-level aggregated metrics

4. Role of AI & Technical Architecture

AI Components Used

- IBM Granite Model: Core LLM for natural language understanding and generation
- Retrieval-Augmented Generation (RAG): Retrieves relevant water consumption data from vector database to ground AI responses in real data
- Anomaly Detection Algorithm: Statistical baseline model flags unusual consumption patterns
- Prompt Engineering: Carefully designed system prompts ensure accurate, context-aware responses
- Entity Extraction: Parses building names, dates, and metrics from user queries

How the RAG Pipeline Works

1. User asks a question (e.g., 'How much water did Block A use this week?')
2. Query is embedded and compared against vector store of campus water data records
3. Top-k relevant records are retrieved (daily usage logs, weather data, historical patterns)
4. IBM Granite model receives: retrieved context + user query + system prompt
5. Model generates a human-readable, data-grounded response with insights
6. If anomaly detected, alert is triggered automatically

Sample Interaction

User: 'How was our water usage this week compared to last week?'

AI: 'This week (June 8-14) Block A used 45,000L, which is 12% lower than last week (51,000L). Block B used 38,000L - a 5% increase. A spike was detected in Block B on June 12 (7,200L vs normal 5,400L), possibly indicating a leak. Shall I alert maintenance?'

Tools & Platforms

- IBM Granite Models - via IBM watsonx.ai
- IBM SkillsBuild learning resources for RAG implementation
- Python for data processing and anomaly detection logic
- Vector database (ChromaDB / FAISS) for RAG retrieval
- Simple web/mobile interface or WhatsApp API for chatbot delivery

5. Responsible AI Considerations

Fairness

The system uses building-level aggregated data, not individual user data, to avoid profiling or discrimination. Recommendations are applied equally across all residential blocks regardless of occupant demographics.

Transparency

Every AI-generated insight includes confidence indicators and a link to the source data. Users can see which data points drove a particular recommendation or alert.

Ethics

No personally identifiable information (PII) is stored or processed. The system is designed to conserve water for the common good, not to penalize or surveil individuals. Water conservation suggestions are always optional and educational.

Privacy

Data retention policy: Aggregated data retained for 12 months for trend analysis. Individual usage data is never collected - only building/zone level metering data. Users can opt out of chatbot interaction at any time.

6. Expected Impact

If Implemented

- 20-30% reduction in water wastage through early leak detection and behavioral feedback
- 15-20% cost savings on campus water utility bills
- Faster leak response: from 2-3 days to under 2 hours with AI-driven alerts
- Increased water literacy and conservation awareness among students and staff
- Scalable model that can be replicated across other campuses and institutions
- Contribution to SDG 6.4 (water-use efficiency) and SDG 11.6 (environmental impact)

Who Benefits and How?

- Students: Reliable water availability, awareness of conservation, gamified engagement
- Administration: Lower costs, data-driven decision making, operational efficiency
- Environment: Reduced water extraction, lower carbon footprint from water treatment
- Local community: Reduced strain on shared water resources

7. Conclusion & Key Message

CampusWater AI demonstrates how AI can be a powerful force for sustainability when applied thoughtfully and responsibly. By combining RAG-powered natural language interfaces with basic anomaly detection, the system makes water management accessible, proactive, and engaging.

This project is not about building the most sophisticated AI system - it is about observing a real problem in our surroundings (water wastage on campus) and applying AI in a targeted, responsible way to create measurable impact.

As the saying goes: 'What gets measured gets managed.' CampusWater AI brings measurement, insight, and action together in one intelligent assistant - turning data into conservation.

"AI can be a force for sustainability, inclusion, and positive change."